NuMicro® Family

Arm® -based Microprocessor

Nuvoton OpenWrt 24.10 Project
User Manual

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1 OVERVIEW

The OpenWrt Project is a Linux operating system targeting embedded devices. Instead of creating a single, static firmware, OpenWrt provides a fully writable filesystem with package management. This Nuvoton OpenWrt implementation is based on OpenWrt 24.10.2. For fundamental concepts and detailed documentation, please refer to the official OpenWrt website at https://openwrt.org/.

1.1 Feature List

The Nuvoton OpenWrt 24.10 release includes the following key components and features:

- Linux Kernel
- Arm-Trusted-Firmware (TF-A)
- OP-TEE OS
- U-boot
- LuCI (Web Interface)
- Python3-Nuwriter (Programming Tool)

2 DEVELOPMENT ENVIRONMENT SETUP

To develop projects within the OpenWrt environment, the following requirements must be met:

- Host System: A Linux distribution (e.g., recent releases of Ubuntu, Debian, Fedora, or CentOS).
- Disk Space: A minimum of 20 GB of free disk space.
- Build Packages: Appropriate dependencies and tools installed on the host system.

Nuvoton provides two methods for building images: Native Linux and Docker.

While building directly on a Linux host is supported, system updates may introduce dependency conflicts that lead to build errors. Therefore, Nuvoton recommends the Docker-based environment. Unlike a native setup, Docker provides a containerized, isolated environment specifically for building images. This ensures consistency and prevents build tools from affecting the host operating system.

2.1 Docker Environment

Docker is an open-source project based on Linux containers. Similar to virtual machines, containers provide an isolated environment but are more portable, resource-efficient, and share the host operating system's kernel. Using Docker is the recommended way to quickly set up the OpenWrt build environment.

2.1.1 Installing Docker (Example for Ubuntu 20.04)

1. **Update the local package list:** Synchronize your local package index with the remote repositories to ensure you have the latest information on available packages.

```
$ sudo apt-get update
```

2. **Install prerequisite packages:** Install the necessary tools that allow apt to safely use packages over HTTPS.

```
$ sudo apt install apt-transport-https ca-certificates curl software-properties-
common
```

3. **Add Docker's official GPG key:** Download and add the GPG key to your system to ensure the authenticity of the Docker packages.

```
$ curl -fsSL https://download.docker.com/linux/ubuntu/gpg | sudo apt-key add -
```

4. **Add the Docker repository to APT sources:** Define the official Docker repository as a source for your package manager.

```
$ sudo add-apt-repository "deb [arch=amd64]
https://download.docker.com/linux/ubuntu focal stable"
```

5. **Refresh the package database:** Run the update command again to include metadata from the newly added Docker repository. This step ensures that the package manager recognizes the Docker packages from the new source.

```
$ sudo apt-get update
```

6. **Install Docker Engine:** Finally, install the Docker Engine, command-line interface, and container runtime components.

```
$ sudo apt-get install docker-ce docker-ce-cli containerd.io
```

2.1.2 Setting Up the Nuvoton Build Environment

Nuvoton provides a set of scripts to automate the Docker image creation and management. You can find the source files at: MA35D1_Docker_Script.

1. **Build the Docker Image:** Run the build script to generate the Docker image and set up the shared folder. The host folder will be mounted at */home/$USER/shared* inside the container.

```
$ ./build.sh
```

2. **Enter the Docker Container:** Use the *join.sh* script to access the container environment. Upon success, the command prompt will change to [user name]@[container id]:~$.

```
$ ./join.sh
ma35d1_test
test@575f27a6d251:~$
```

3. **Create a Working Directory:** Inside the container, create a dedicated folder within the shared directory to store the OpenWrt source code. This ensures your data persists on the host machine.

```
test@575f27a6d251:~$ mkdir shared/openwrt
test@575f27a6d251:~$ cd shared/openwrt
```

4. **Configure Git Identity:** Before using Git for the first time, you must configure your global identity. These settings are required for Git to function properly during the build and download process.

```
test@575f27a6d251:~$ git config --global user.email "test@test.test.test"
test@575f27a6d251:~$ git config --global user.name "test"
```

5. **Download the OpenWrt Source Code:** Clone the Nuvoton OpenWrt project repository from GitHub to your working directory.

```
test@575f27a6d251:~$ git clone https://github.com/OpenNuvoton/Nuvoton-OpenWrt-24.10.git
```

For further information on Docker, please refer to the Official Docker Documentation.

2.2 Native Linux Environment

If you choose a native Linux environment for building images, several prerequisite packages must be installed before starting the OpenWrt project. For a comprehensive list of dependencies for other Linux distributions, please refer to the OpenWrt Build System Documentation.

1. **Install OpenWrt Dependencies (Ubuntu 22.04):** Run the following command to install the required packages for the OpenWrt project:

```
$ sudo apt install build-essential gawk gcc-multilib flex git gettext libncurses5-dev libssl-dev python3-distutils rsync unzip zlib1g-dev
```

2. **Install OpenWrt Dependencies (Ubuntu 20.04 and older):** For older Ubuntu releases, use the following command:

```
$ sudo apt install build-essential ccache ecj fastjar file g++ gawk \
gettext git java-propose-classpath libelf-dev libncurses5-dev \
libncursesw5-dev libssl-dev python python2.7-dev python3 unzip wget \
python-distutils-extra python3-setuptools python3-dev rsync subversion \
swig time xsltproc zlib1g-dev
```

3. **Install Component-Specific Dependencies:** In addition to the core OpenWrt packages, sub-projects require specific tools:
   - For Arm Trusted Firmware (TF-A):

```
$ sudo apt install libssl-dev device-tree-compiler
```

- For OP-TEE OS:

```
$ pip3 install pycryptodomex pyelftools
```

- For Python3-Nuwriter:

```
$ pip3 install pyusb usb crypto ecdsa crcmod tqdm pycryptodome
```

4. **Download the OpenWrt Source Code:** Once the environment setup is complete, clone the Nuvoton OpenWrt project repository:

```
$ git clone https://github.com/OpenNuvoton/Nuvoton-OpenWrt-24.10.git
```

3 BUILD IMAGE

This section provides detailed instructions and the necessary process for building a firmware image for the Nuvoton MA35 series.

3.1 Update Feeds Scripts

1. **Update and Install Feeds:** After downloading the OpenWrt source code, execute the following commands to update and install the OpenWrt feeds:

```
$ ./scripts/feeds update -a
$ ./scripts/feeds install -a
```

2. **Handling SSL Errors:** If you encounter a gnutls_handshake() failed error (usually due to missing CA certificates on the host machine), you can bypass the SSL verification by running the following command before retrying the update:

```
$ export GIT_SSL_NO_VERIFY=1
```

3.2 OpenWrt Configurations

The Nuvoton OpenWrt project supports various platforms, including the MA35D1 (IoT and SOM), MA35D0 (IoT), and MA35H0 (HMI). Ensure you use the correct configuration for your target board.

1. **Apply Default Configuration:** For example, to configure the environment for the MA35D1 IoT board, copy the provided configuration file to *.config*:

```
$ cp Nuvoton/config/config_ma35d1_iot .config
```

2. **Enter Menu Configuration:** Run the following command to open the configuration menu and verify the settings:

```
$ make menuconfig
```

3. **Select Target Parameters:** As shown in Figure 3-1, confirm that the Target System is "Nuvoton MA35D1", the Subtarget is "MA35D1 IoT", and the Profile matches your specific memory size and boot storage.

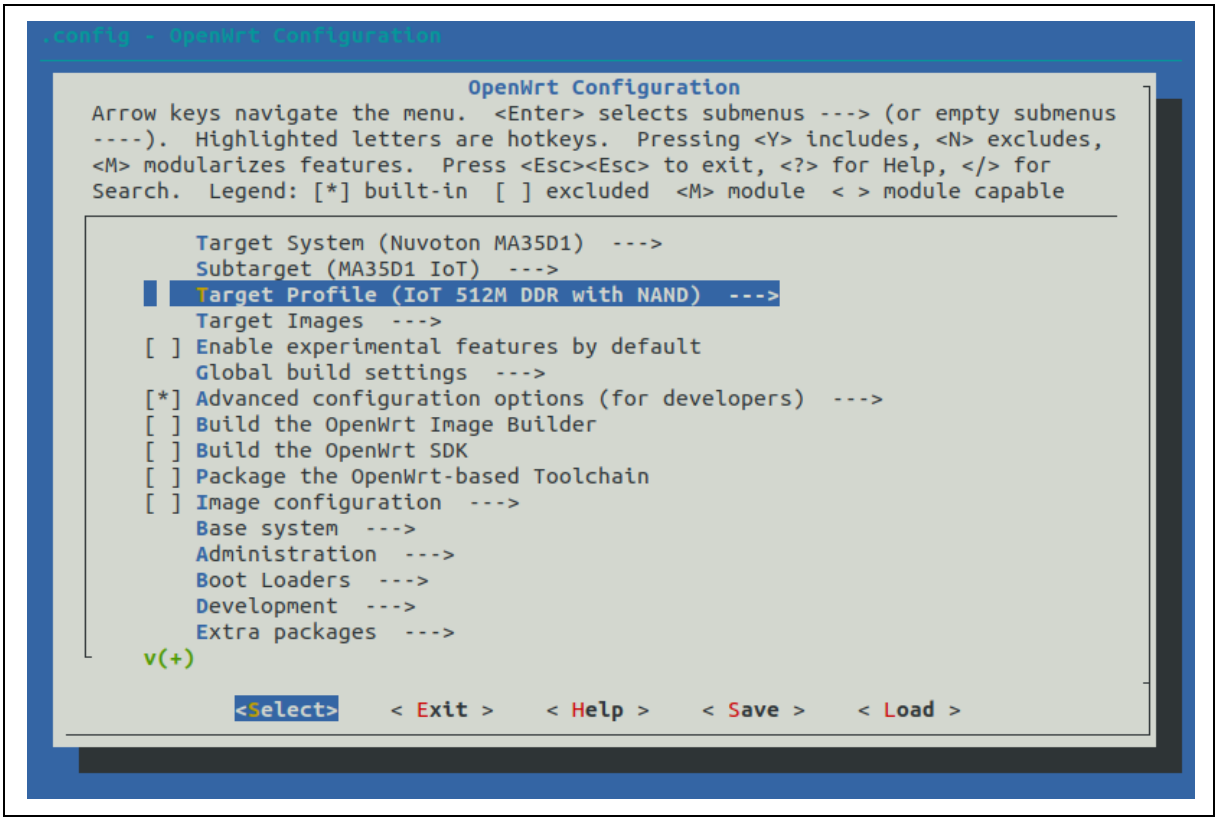

Figure 3-1 Select Target, Subtarget and Profile for MA35D1

4. **Configure Kernel Repository:** In the “Advanced configuration options” page, specify the Git repository and the branch/commit for the Linux kernel source (see Figure 3-2). The default setting uses HTTPS for GitHub. If needed, you can switch to other repository managers as listed in Table 3-1.

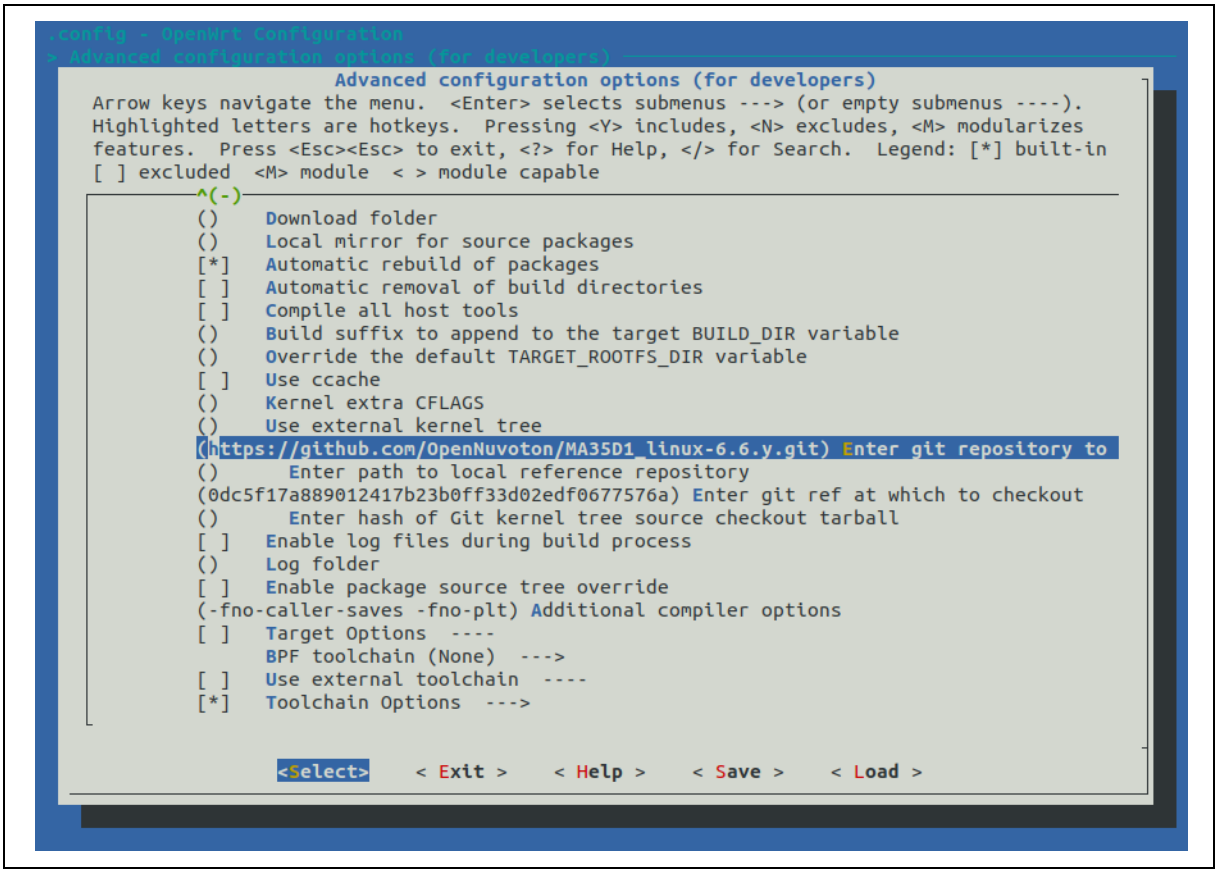

Figure 3-2 Configure Kernel Repository for MA35D1

Repository Manger	URL
Github	https://github.com/OpenNuvoton/MA35D1_linux-6.6.y.git
Gitlab	https://gitlab.com/OpenNuvoton/Cortex-A-Family/MA35D1_linux-6-6-y.git
Gitee	https://gitee.com/OpenNuvoton/MA35D1_linux-6.6.y.git

Table 3-1 Linux Kernel Repositories for MA35D1

5. **Boot Loader Settings:** On the “Boot Loaders” page, the OP-TEE, TF-A, and U-Boot options should be enabled, as shown in Figure 3-3. Essential hardware initialization parameters—such as CPU speed, PMIC configuration, and DDR settings—are defined specifically within the TF-A options to ensure the system initializes and boots correctly.

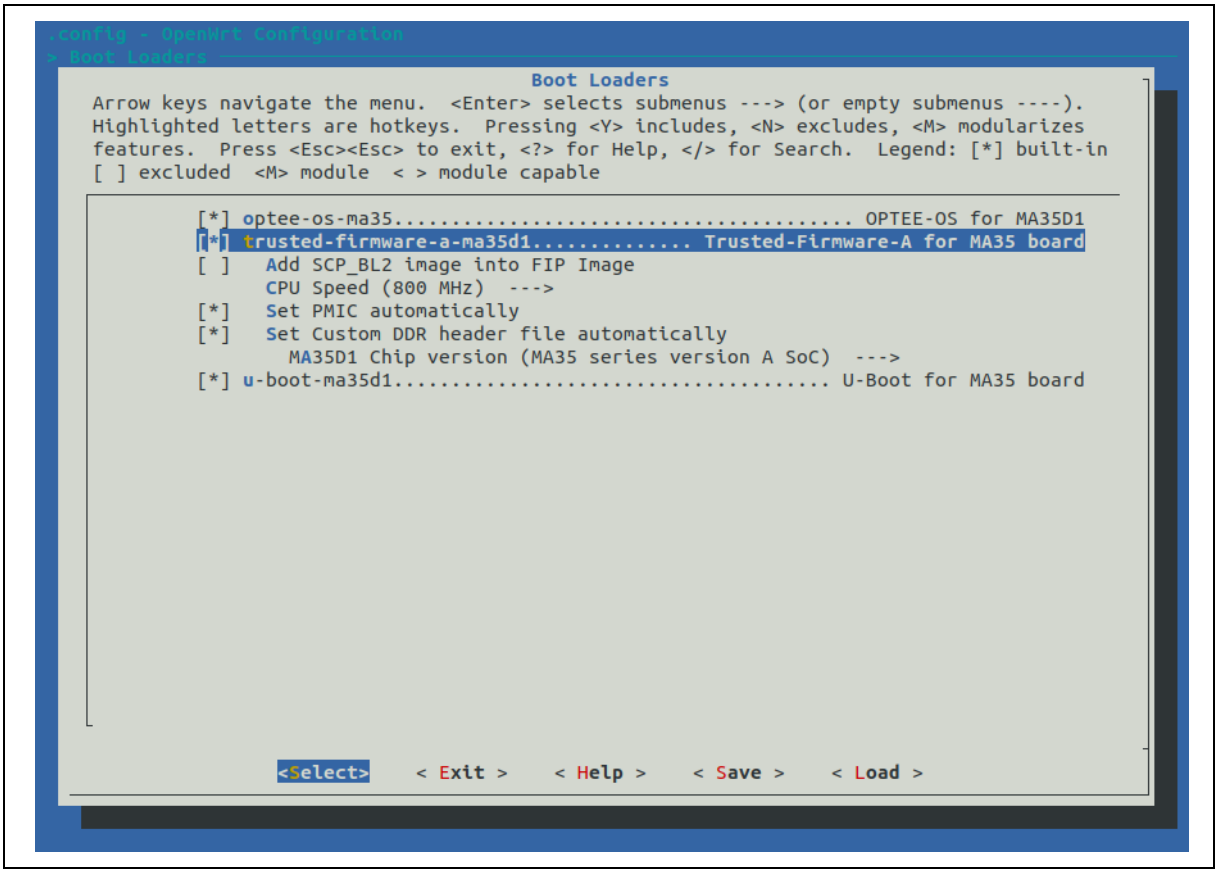

Figure 3-3 Boot Loaders Configurations for MA35D1

3.3 Linux Kernel Configurations

1. **Enter Kernel Menuconfig:** After completing the OpenWrt configuration, run the following command to configure the Linux kernel:

```
$ make kernel_menuconfig
```

2. **Toolchain and Source Download:** Before the kernel configuration menu appears, the system will build the required toolchain and download the MA35D1 Linux BSP. This process may take approximately 90 minutes depending on your hardware. Once the menu appears, you can save and exit directly.
3. **Configuration Storage:** If you modify any kernel settings, the configuration for the MA35D1 IoT board will be saved at: *target/linux/ma35d1/iot/config-6.6*.

3.4 Build Image

3.4.1 Output Images

1. **Run the Build Command:** Once the configuration is complete, run the make command to start the build process. A clean build typically takes around 30 minutes depending on your system performance.

```
$ make
```

2. **Locate Generated Images:** For the MA35D1 IoT board, the generated images are located in the *bin/targets/ma35d1/iot/* directory. The primary output files are described in Table 3-2.

Image Name	Description
openwrt-ma35d1-${Subtarget}-${Profile}-pack.bin	OpenWrt pack image using NAND / SPINAND / SDCARD / eMMC booting, can be used by MA35D1 nuwriter tool.
openwrt-ma35d1-${Subtarget}-${Profile}-squashfs-firmware.bin	OpenWrt firmware image using NAND / SPINAND booting, includes Linux kernel uImage, and OverlayFS root filesystem.
openwrt-ma35d1-${Subtarget}-${Profile}-squashfs-sysupgrade.bin	OpenWrt system upgrade image using NAND / SPINAND booting.
openwrt-ma35d1-${Subtarget}-${Profile}-ext4-sysupgrade.gz	OpenWrt system upgrade image using SDCARD / eMMC booting.

Table 3-2 MA35D1 OpenWrt Generated Images in Output Directory

3. **Rebuilding Loaders:** If you change the Target System, Subtarget, or Target Profile settings, you must clean the loader packages before rebuilding to ensure the new settings are applied.

```
$ make package/boot/arm-trusted-firmware-ma35d1/clean
$ make package/boot/optee-os-ma35d1/clean
$ make package/boot/uboot-ma35d1/clean
```

3.4.2 Error Troubleshooting

1. **Collect Detailed Error Logs:** If a compilation error occurs, use the following command to run a single-threaded build and collect detailed logs for analysis:

```
$ make -j1 V=sc
```

3.5 Deploy Image

Nuvoton provides a "pack image" that integrates TF-A, OP-TEE, U-Boot, Linux kernel, and the filesystem for various booting storages. You can use the NuWriter GUI or command-line tool to program these images.

3.5.1 NuWriter GUI Tool

1. **Attach the Board:** Open the NuWriter GUI, choose the correct DDR image for your hardware, and click the “Attach” button to connect to the MA35D1 demo board (as shown in Figure 3-4).

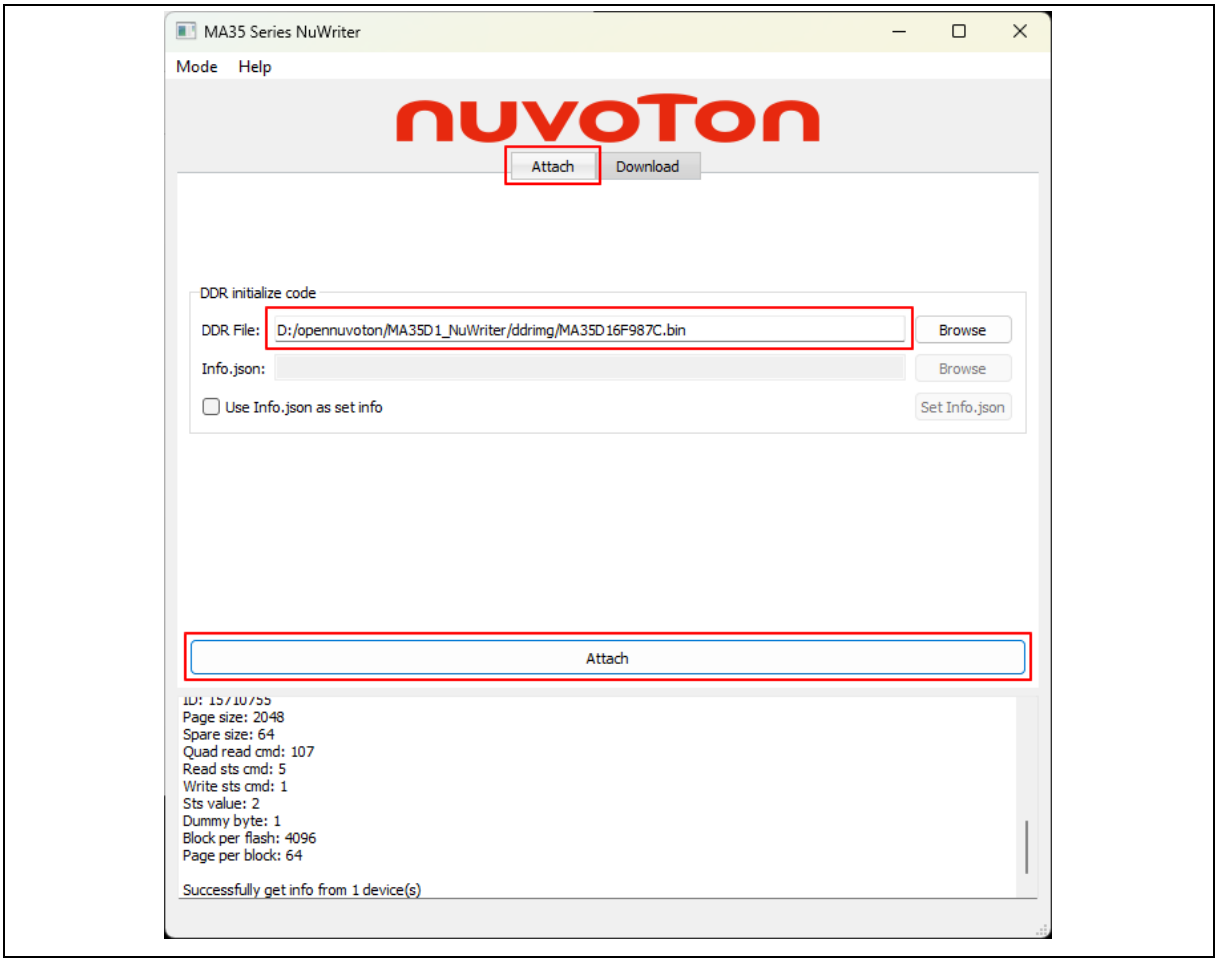

Figure 3-4 Use NuWriter to Attach MA35D1 IoT Board

2. **Erase Storage:** Select the “Download” tab and specify your target booting storage. Before programming NAND or SPINAND, you must erase the storage first (as shown in Figure 3-5)

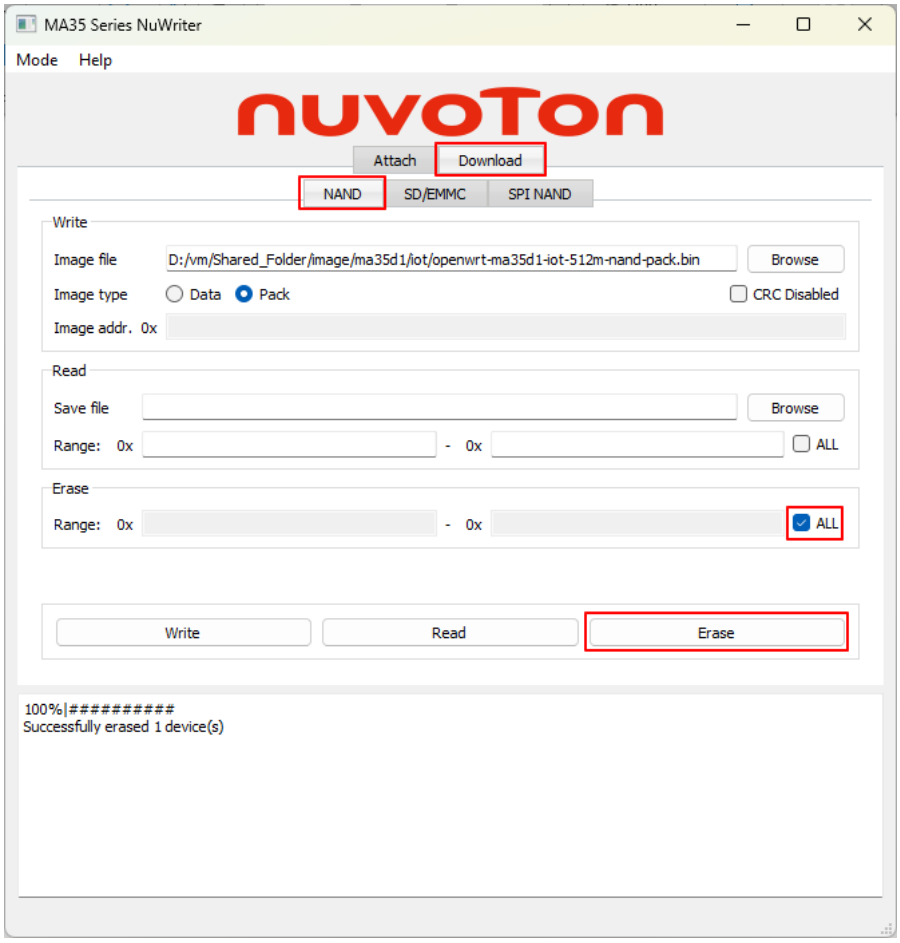

Figure 3-5 Use NuWriter to Erase MA35D1 IoT Board

3. **Program the Pack Image:** Select the correct pack image file, set the image type to “Pack”, and click the “Write” button to start programming (as shown in Figure 3-6).

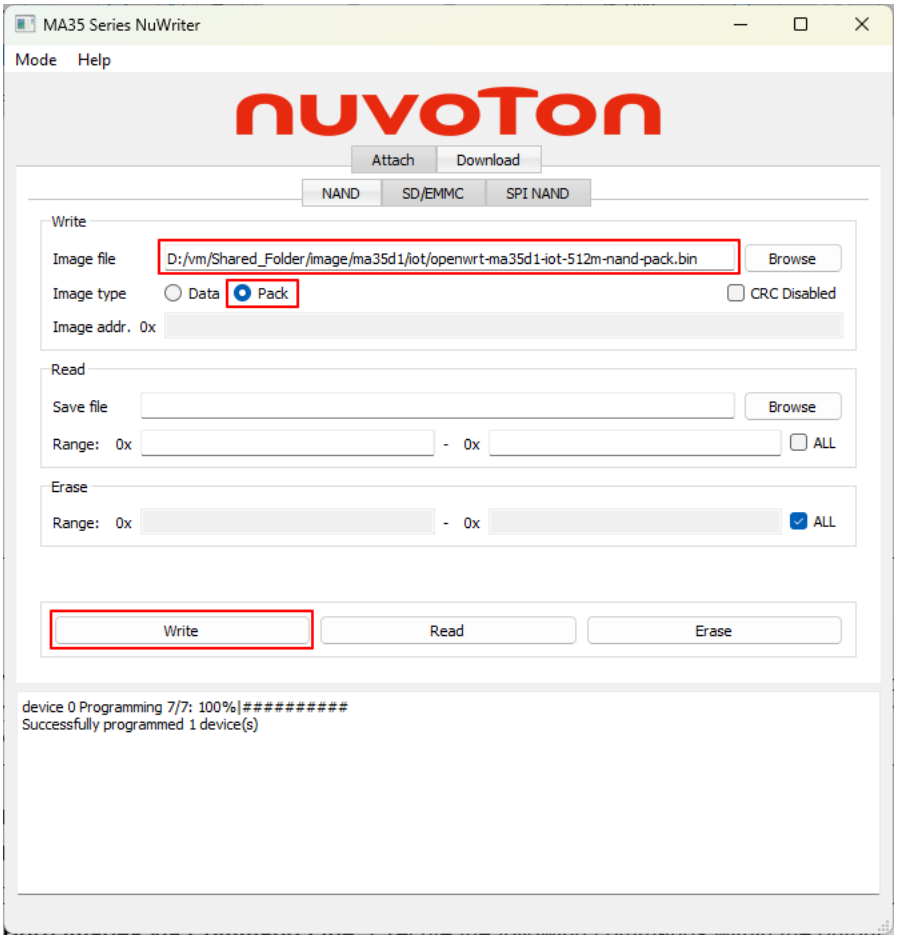

Figure 3-6 Use NuWriter to Program MA35D1 IoT Board

3.5.2 NuWriter Command-Line Tool

1. **Verify Connection:** Use the lsusb command to verify that the board is correctly connected to your computer. **Note:** The nuwriter command may not function properly within a Docker environment due to USB access limitations.

```
$ sudo lsusb
Bus 001 Device 002: ID 80ee:0021 VirtualBox USB Tablet
Bus 001 Device 001: ID 1d6b:0001 Linux Foundation 1.1 root hub
/OpenWrt/bin/targets/ma35d1/iot$
```

2. **Burn Images via Command Line:** Execute the following commands within the output directory (*bin/targets/ma35d1/iot/*) to program different storage types:

- For NAND Flash:

```
/OpenWrt/bin/targets/ma35d1/iot$ nuwriter.py -a enc_ddr3_winbond_512mb.bin
/OpenWrt/bin/targets/ma35d1/iot$ nuwriter.py -e nand all
/OpenWrt/bin/targets/ma35d1/iot$ nuwriter.py -w nand openwrt-ma35d1-iot-512m-nand-pack.bin
```

- For SPINAND Flash:

```
/OpenWrt/bin/targets/ma35d1/iot$ nuwriter.py -a enc_ddr3_winbond_512mb.bin
```

```
/OpenWrt/bin/targets/ma35d1/iot$ nuwriter.py -e spinand all
/OpenWrt/bin/targets/ma35d1/iot$ nuwriter.py -w spinand openwrt-ma35d1-iot-512m-spinand-pack.bin
```

- For SDCARD:

```
/OpenWrt/bin/targets/ma35d1/iot$ nuwriter.py -a enc_ddr3_winbond_512mb.bin
/OpenWrt/bin/targets/ma35d1/iot$ nuwriter.py -w sd openwrt-ma35d1-iot-512m-sdcard1-pack.bin
```

3.6 Secure Boot

This section describes how to enable and configure the Secure Boot feature for the MA35D1/MA35D0/MA35H0 platforms.

3.6.1 Configuration

1. **Enable Secure Boot:** Navigate to the "Target Images" menu to enable the Secure Boot feature. Here, you can also customize the AES key and ECDSA key settings as shown in Figure 3-7.

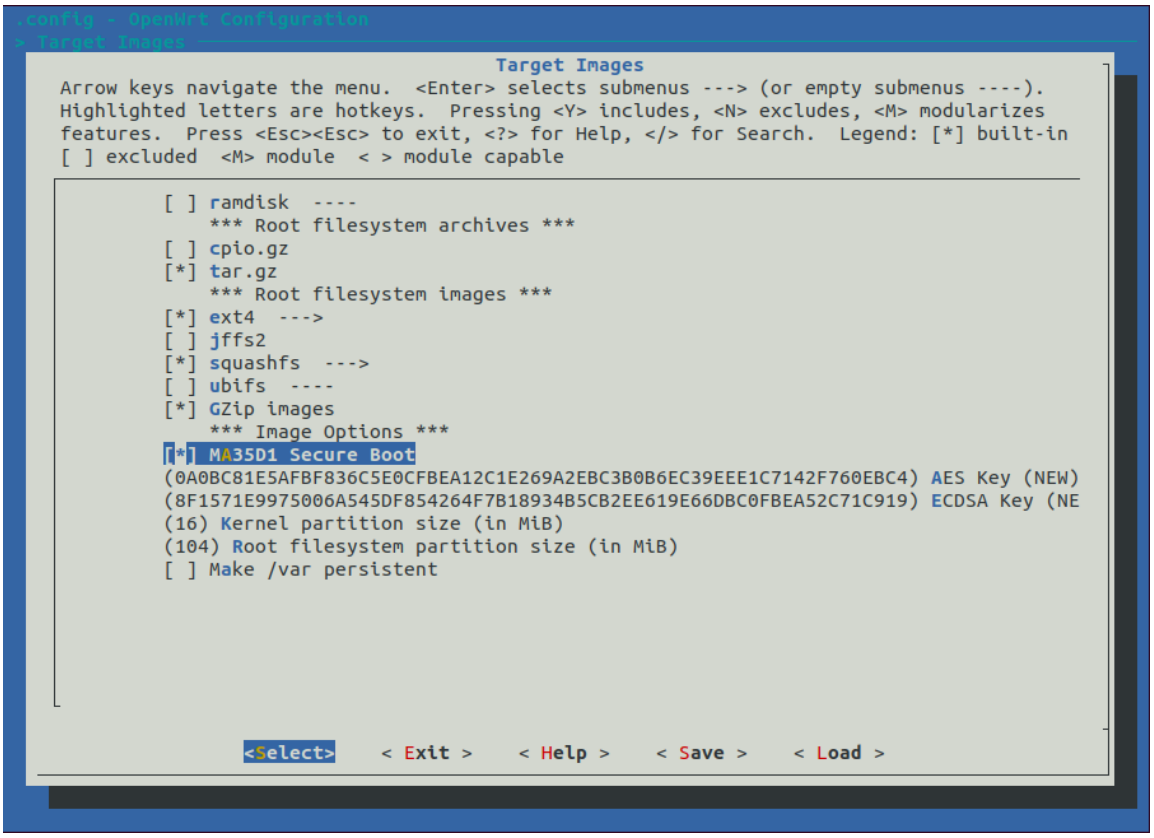

Figure 3-7 Enable Secure Boot feature

3.6.2 Build Image

1. **Rebuild Security Components:** Whenever you modify Secure Boot settings, you must clean the TF-A, OP-TEE, and U-Boot packages to ensure the new keys and security configurations are applied during the next build:

```
$ make package/boot/arm-trusted-firmware-ma35d1/clean
```

```
$ make package/boot/optee-os-ma35d1/clean
$ make package/boot/uboot-ma35d1/clean
```

2. **Identify Encrypted Output:** Once the build process is complete, the encrypted images will be generated with an "-enc" suffix (e.g., ...-pack-enc.bin), distinguishing them from standard images.

```
openwrt-ma35d1-iot-512m-nand-pack-enc.bin
openwrt-ma35d1-iot-512m-nand-squashfs-sysupgrade-enc.bin
openwrt-ma35d1-otp-key-enc.json
```

3. **OTP Key Generation:** In addition to the encrypted firmware, an OTP (One-Time Programmable) key file in JSON format will be generated (e.g., openwrt-ma35d1-otp-key-enc.json). This file contains the security parameters required for the device:

```
$ cat openwrt-ma35d1-otp-key-enc.json
{
  "publicx": "72F84F681092E3A05C1437E3E40534962A5C70556025D348FF9DB97D6AF83EB5",
  "publicy": "8D32DAC7AB6F90332E8E0060E159E0B31502BB4FB2D78369F02D1F5B0C335AD3",
  "aeskey": "0A0BC81E5AFBF836C5E0CFBEA12C1E269A2EBC3B0B6EC39EEE1C7142F760EBC4"
}
```

3.6.3 Deploy OTP Key File

For Secure Boot to function, both the encrypted pack image and the OTP key file must be programmed into the MA35D1. The key file is written to the device's internal Key Store. For detailed security theory, please refer to the *MA35D1 Secure Boot Application Note*.

1. **NuWriter GUI Method:** Switch the NuWriter tool to “OTP mode”, choose the generated OTP key file, and click the “Write” button to program the Key Store (as shown in Figure 3-8).

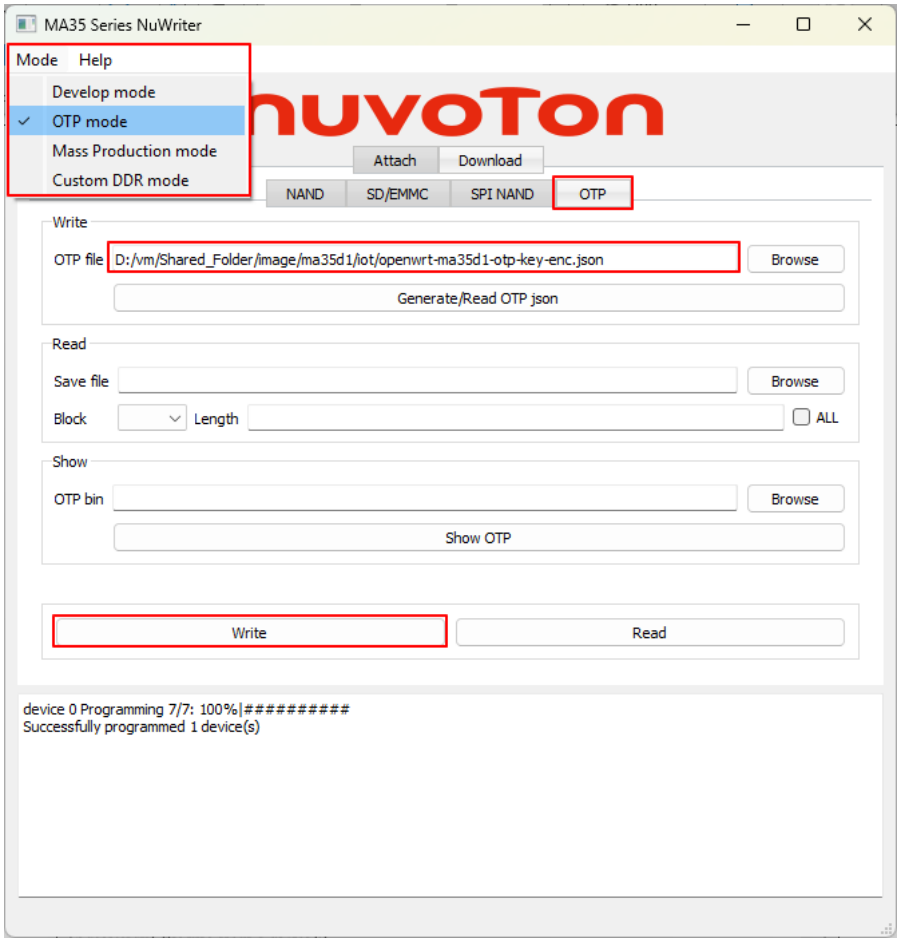

Figure 3-8 Use NuWriter to Program MA35D1 OTP key

2. **NuWriter Command-Line Method:** Use the following command to program the OTP key file via the terminal:

```
/OpenWrt/bin/targets/ma35d1/iot$ nuwriter.py -w otp openwrt-ma35d1-otp-key-enc.json
```

3.7 emWin Support (TODO)

This section describes how to enable the emWin graphic library on supported Nuvoton platforms. Currently, the following development boards equipped with display panels support emWin:

NuMaker-HMI-MA35D1

NuMaker-HMI-MA35H0

3.7.1 Copy the emWin Source to OpenWrt

1. **Extract the Source Package:** After downloading the emWin package from the Nuvoton website, locate the Linux emWin tarball. For the MA35D1, execute the following commands to extract the source:

```
$ cd package/nuvoton/emwin-ma35d1/
$ tar zxvf MA35D1_Linux_emWin_Package_20240110.tar.gz
```

2. **Rename the Directory:** The extracted directory includes a date suffix by default. You must

remove this suffix to ensure the OpenWrt build system can correctly locate the source files:

```
mv MA35D1_Linux_emWin_Package_20240110 MA35D1_Linux_emWin_Package
```

3. **Version Requirement:** Please note that OpenWrt only supports emWin package versions released after the year 2024.

3.7.2 Configuration

1. **Enable Features in Menuconfig:** Navigate to the “Nuvoton” menu in menuconfig. From there, you can enable the emWin and tslib (touchscreen library) features as shown in Figure 3-9.

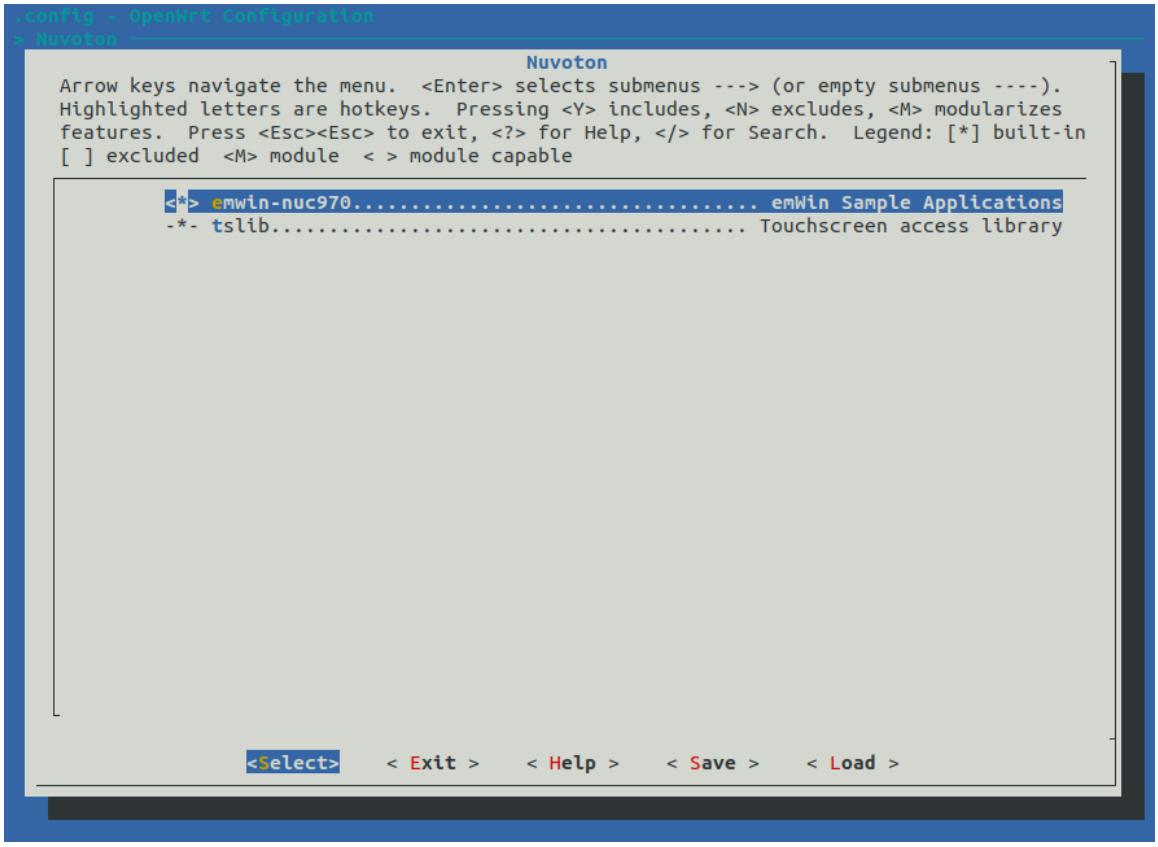

Figure 3-9 Enable emWin and tslib features

3.7.3 Samples

1. **Explore Samples:** The emWin source package includes several demonstration projects located under the *Sample* folder:

```
$ ls package/nuvoton/emwin-ma35d1/MA35D1_Linux_emWin_Package/Sample/
GUIDemo  SimpleDemo  SimpleDemoAppWizard
```

2. **Build and Execute:** During the image-building process, these samples are automatically compiled and copied to the root filesystem (rootfs). Once OpenWrt boots up, you can run the emWin demos from the target terminal:

```
root@OpenWrt:/# ls usr/bin/*Demo*
usr/bin/GUIDemo
usr/bin/SimpleDemoAppWizard
```

```
usr/bin/SimpleDemo
```

4 TEST OPENWRT

This chapter describes how to verify the OpenWrt system, configure network settings, and perform firmware updates.

4.1 Booting Messages

Upon powering on the device, the console will display the OpenWrt boot logs as shown in Figure 4-1. These messages verify that the kernel and root filesystem have initialized correctly.

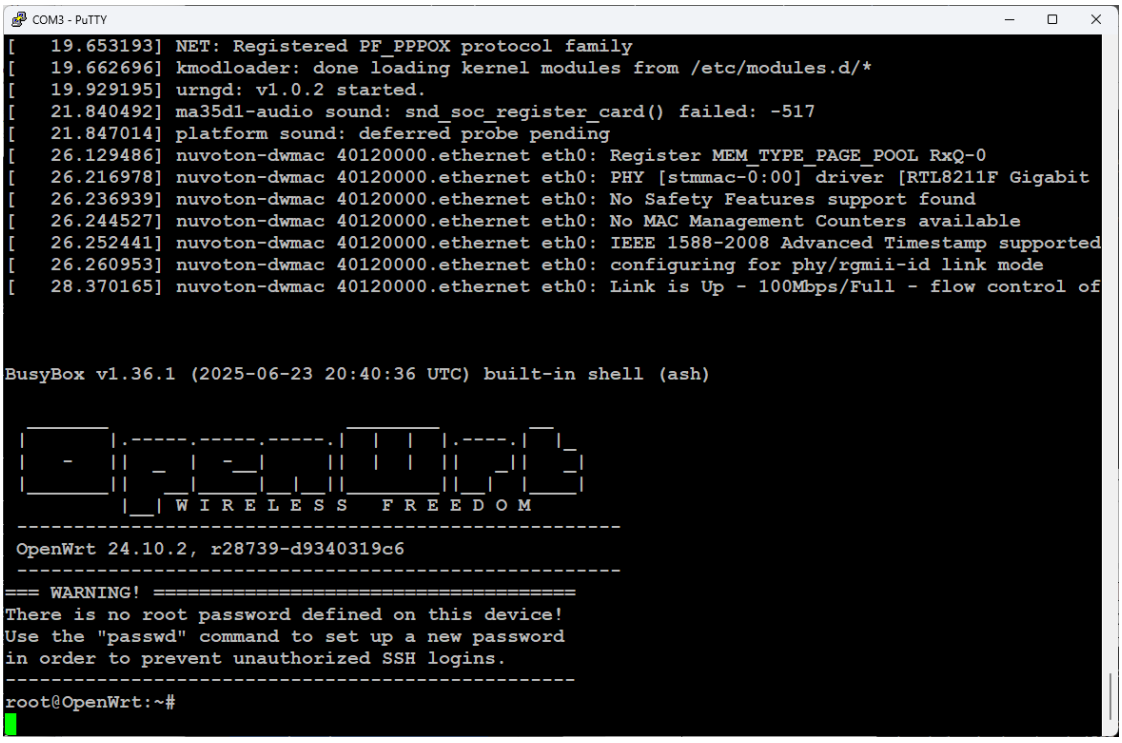

Figure 4-1 OpenWrt Booting Messages

4.2 Network Settings

To check the current network settings, execute the following command:

```
uci show network
```

By default, the system is configured to obtain an IP address via DHCP, as shown in Figure 4-2.

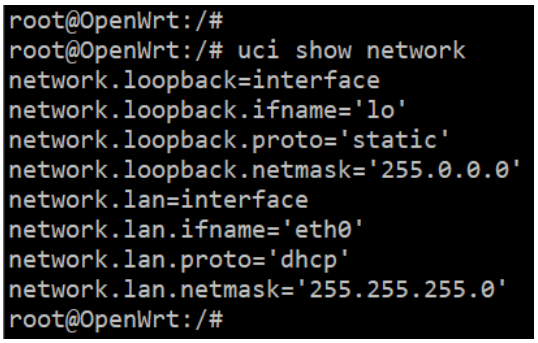

Figure 4-2 Default Network Settings

To change the network settings, user can modify the file path /etc/config/network directly, or run the “uci set” command. For example, to change to a static address, user can run following commands to modify and reset the network settings.

You can change network settings by manually editing */etc/config/network* or by using the uci utility. For example, to switch to a static IP address, run the following commands:

```
uci set network.lan.proto=static
uci set network.lan.ipaddr=192.168.1.100
uci set network.lan.netmask=255.255.255.0
/etc/init.d/network restart
```

4.3 LuCI Web Interface

To access the LuCI interface, open a web browser (e.g., Chrome) and navigate to https://<YOUR_IP_ADDRESS>. The default configuration has no root password. You can log in directly by clicking the “Login” button, as shown in Figure 4-3.

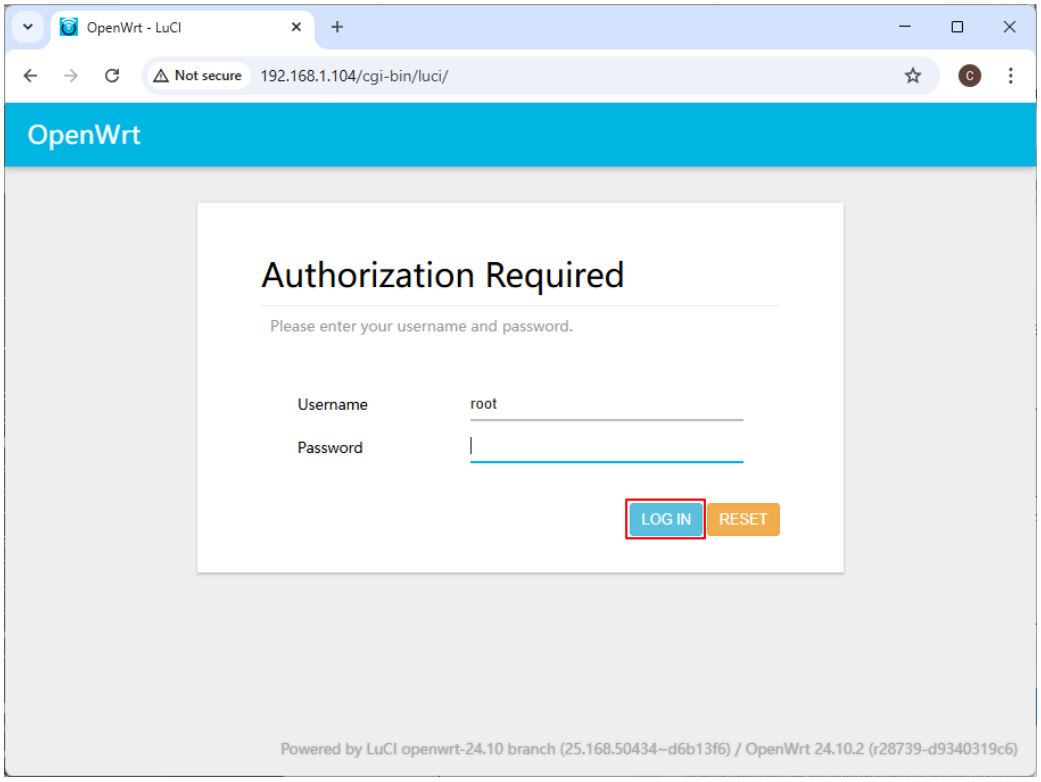

Figure 4-3 Login Screen of LuCI

After logging in, the Status Overview page displays system details for the Nuvoton MA35D1-IoT, as shown in Figure 4-4. It lists the firmware and kernel versions, while a yellow banner warns that no root password has been set.

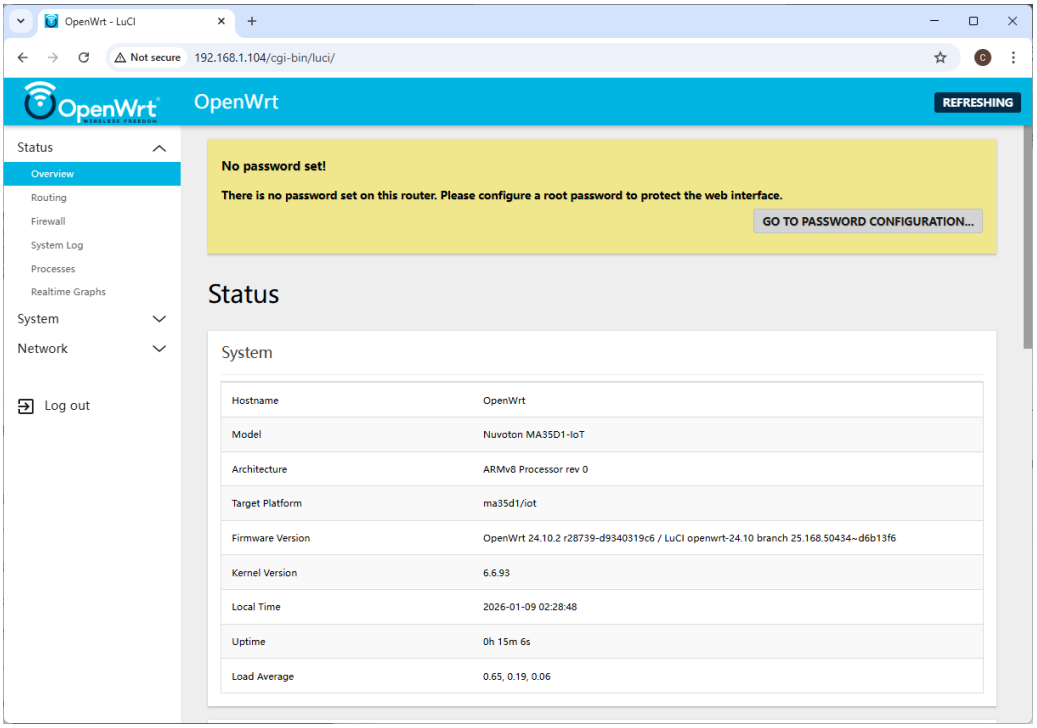

Figure 4-4 Overview Screen of LuCI

4.4 Firmware Update

The LuCI interface allows you to upgrade the firmware image, which includes the Linux kernel and the OverlayFS root filesystem.

For the MA35D1 IoT board, ensure you use the appropriate sysupgrade file matching your specific storage type:

```
openwrt-ma35d1-iot-512m-nand-squashfs-sysupgrade.bin
openwrt-ma35d1-iot-512m-spinand-squashfs-sysupgrade.bin
openwrt-ma35d1-iot-512m-sdcard1-ext4-sysupgrade.gz
```

Navigate to “System -> Backup / Flash Firmware”. In the "Flash new firmware image" section, click "FLASH IMAGE", upload your sysupgrade file, and follow the on-screen prompts (as shown in Figure 4-5).

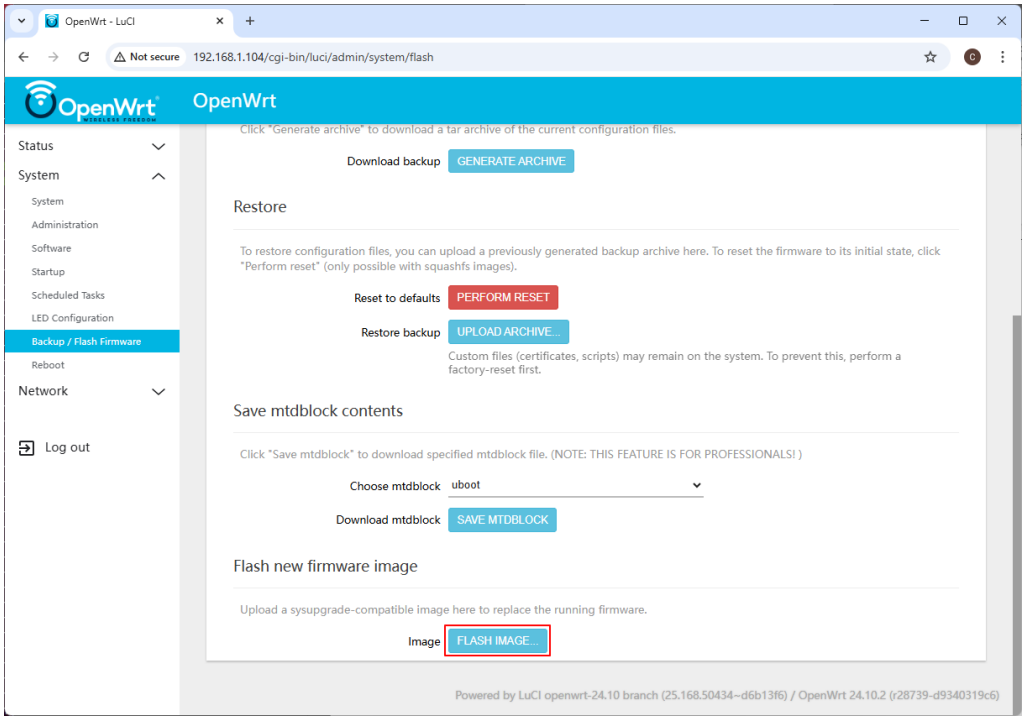

Figure 4-5 Firmware Upgrade in LuCI

5 REVISION HISTORY

Date	Revision	Description
2026.01.09	1.00	● Initial version. ● Support MA35D1 platform.

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